SMU
SINGAPORE MANAGEMENT UNIVERSITY

School of
Computing and
Information Systems

IS459 Big Data Architecture

PROJECT

Format & Deliverables

Graded components

- Project proposal presentation (10 mins) in Week 9 : **5%**
  - Slides (PDF or PPTX)

- Final project : **30%**

  1. Final project presentation (20 mins) in Week 13: **15%**

  2. Final project artefacts: **15%**
     - Slides (PDF or PPTX)
     - Video snippets of your project architecture/pipeline
     - Any other items (deployment, GitHub repository, source code etc.)

Dataset

- Use the following dataset for your project implementation.

  IBM Transactions for Anti Money Laundering (AML)

- Sign into Kaggle and download the dataset

- Inspect the data

Data Dictionary

Attribute	Description
Timestamp	Date and time when the transaction occurred
From Bank	Identifier of the bank from which the transaction originates
Account	Identifier of the **sending/originating account**
To Bank	Identifier of the bank receiving the transaction
Account.1	Identifier of the **receiving/destination account**
Amount Received	Amount received by the destination account
Receiving Currency	Currency in which the amount is received
Amount Paid	Amount paid by the originating account
Payment Currency	Currency in which the payment is made
Payment Format	Method/format used for the transaction, e.g. Cheque, Credit Card, ACH, Wire, etc.
Is Laundering	Binary label: **1 = laundering transaction;** **0 = legitimate transaction**

Instructions

- Come up with **2 business questions** that can be answered by processing and analyzing this dataset. The 2 questions must be *distinct/unique* (cannot be similar).

- For **each business question**, identify and document the following:
  What is the business question?
  Who will benefit from answering this business question and why?

- For your business questions, should you wish to *augment* the Kaggle dataset with additional data, you are more than welcome to do so as long as this *addition* supports answering the business questions. Please clearly identify additional data sources by **citing** them (e.g. provide the URL/link to the data sources).

Instructions

Please construct one or more big data pipeline(s) to answer the 2 business questions *(see previous slide)*.

1. Use one or more AWS service(s) *(you may use other services that we did not explore in class)* to construct one or more **end-to-end big data pipeline(s)**.
   Please **DO NOT purchase anything from AWS Marketplace**.

2. For each data pipeline, create an architecture diagram.

3. The pipeline should start with **data ingestion** and end with **insight visualization**.

4. You can use the AWS account shared with you for your project implementation. **Budget is 250 USD.**

5. Please avoid expensive services such as Amazon Redshift, Amazon RDS, and Amazon QuickSight. Use cost-effective alternatives whenever possible.

Project Deliverables & Submission

- **Graded components**
  - Project proposal presentation (10 mins): **5%** **- Upload to eLearn before your Week 9 lesson time**
    - Slides (PDF or PPTX)
  - Final project presentation (20 mins): **15%** **- Presentation in Week 13**
    - **Slides** (PDF or PPTX) **- Upload to eLearn before your Week 13 lesson time**
  - Final project artefacts: **15%** **- Upload to eLearn & put links/info in Presentation slides before your Week 13 lesson time**

    - **Video snippets** of your project architecture and big data pipeline
      - A video must show (with narration or annotations) how to set up the pipeline end-to-end.
      - A video must show (with narration or annotations) how to run the pipeline end-to-end.
      - *Upload to YouTube. Put links to the video(s) in Presentation Slides (appendix)*
    - **Source code files** *(e.g ETL scripts, analysis scripts) &* **Any other items**
      - *Upload to eLearn or put links/info in Presentation Slides (appendix)*

Week 9 :: Proposal Pitch

- Duration
  - 10 minutes pitch / group
  - 5 minutes feedback *(by the teaching team)*

- You (as a group) must present (using slides / drawing if needed):
  - What are the business questions you've identified?
  - Who will **benefit** from what you're building?
    - How will the **intended audience** benefit from your big data pipeline and output?
    - What pain points (if any) will your solution solve for the **intended audience**?
  - Describe the proposed **end-to-end big data pipeline**:
    - Are you using any additional data?
    - Where do you plan to **store** (if any) these data?
    - What **processing** do you plan to do? *(if any)*
    - What **analyses** do you plan to do? What is the expected output from the analyses?
    - If you have a *rough/draft* architecture diagram, please show.

- You can select 1-2 people from your group to pitch *(due to time constraint)*

- ALL members **must be present *(stand in front of the classroom together)*** during the in-class/F2F pitch.
  - Missing members (or late arrivals) will be penalized with ZERO (0) mark for the proposal grade.
  - All members must be ready to answer questions from the teaching team and provide input.

- You can adjust and fine-tune your project direction/details later, based on feedback from the teaching team.

Week 13 :: Final Presentation

- Duration
  - 15 minutes presentation
  - 5 minutes Q&A *(with the teaching team + audience)*
- You (as a group) must present:
  - **Slides**
    - Quickly and briefly remind us what are the main business questions, who is the main audience, what is the main objective, big data pipeline architecture and tools/packages/libraries used in your solution development
    - **Individual Contributions -** Include a slide clearly stating what **each team member contributed** to the project (e.g., data ingestion, ETL, Spark processing, analysis, visualization etc.).
  - **Big Data Pipeline execution demo**
    - **[Option 1]** You can choose to do **live demo** *(You need to plan this carefully - think about the time taken to run the whole flow)*
    - **[Option 2]** Prepare one or more **video clips**, play them, and **narrate live**
- ALL members **must be present** ***(stand in front of the classroom together)*** during the presentation.
  - Missing members (or late arrivals) will be penalized with ZERO (0) mark for the final project presentation grade.
  - **Every member** must **present/demo** at least **one component/feature** of the big data pipeline.
  - All members must be ready to answer questions from the teaching team and audience.

AWS Resource Cleanup: After your final presentation, you must delete all AWS resources created for the project to prevent further charges.

Project Evaluation - What Are We Looking For?

Both Business Value and Technical Solution are important for a strong project.

Business Value

- Are the business questions meaningful?
- Who benefits from the solution, and how?
- Do the resulting insights provide useful/actionable value?

Technical Solution

- Is the big data pipeline appropriately designed?
- Is it correctly implemented and working?
- Does the pipeline effectively support answering the business questions?

A sophisticated pipeline alone does not guarantee a strong project. The solution must also address a meaningful business problem and deliver useful insights.

X-Factor - Going Beyond the Labs

An **X-factor** is a meaningful enhancement that goes beyond what was directly demonstrated in the course labs.

Examples include:

- Pipeline automation or orchestration using tools such as Airflow or AWS Step Functions.
- Streaming / near-real-time processing
- Cost or performance optimization with supporting evidence
- Meaningful integration of additional data sources
- Advanced interactive visualization beyond what was demonstrated in the labs

Simply adding complexity does not constitute an X-factor. The enhancement should have a clear purpose and add value to the solution.

Use of ML/AI is optional. If used, it should meaningfully contribute to answering the business questions. **Simply adding an ML/AI component does not automatically make the solution stronger or qualify as an X-factor.**

Grading Rubrics :: Presentation

Marks	Details
C+ & below	**Presentation with Demo is sloppy.** Difficult to understand how the solution works, how it helps solve the business problems (or answer the business questions), or clarity of technologies used.
B- to B+	**Presentation with Demo is of reasonable standard.** Relatively clear presentation & demo that shows how the solution works, how it helps solve the business problems (or answer the business questions), or clarity of technologies used. Might have some minor non-critical problems here and there.
A- to A	**Presentation with Demo is good.** Clear presentation & demo that is quite seamless. Easy to understand and make sense of what they did and how they did it.
A+	**Presentation with Demo is exceptional.** Crystal clear presentation & demo that is seamless. No clarification questions needed to make sense of what they did. Must also demonstrate at least one meaningful X-factor to qualify for an A+.

Grading Rubrics :: Solution Itself *(Big Data Pipeline)*

Problem(s) solving and solution appropriateness

Marks	Details
C+ & below	**Problem is not solved and/or not a useful solution.**
B- to B+	**Problem is somewhat solved and is relatively useful.** Might not be completely solved but mostly there.
A- to A	**Problem solved is very useful.** A clear problem that is solved with a decent solution.
A+	**Challenging problem is solved and very useful.** It is a challenging problem to begin with and the solution is very appropriate.

Grading Rubrics :: Solution Itself *(Big Data Pipeline)*

Working big data pipeline and its components/parts, correctly implemented, usable solution

Marks	Details
C+ & below	**Pipeline is somewhat working to not working.** Intermittently working, some parts working, or completely not working.
B- to B+	**Pipeline is buggy with some working parts.** With bugs but nothing critical or show stopping. Pipeline is roughly usable without having to figure out how to use it too much.
A- to A	**Pipeline is working well, and usability is good.** Some minor bugs and with the pipeline relatively intuitive to use.
A+	**Pipeline is working perfectly with great usability.** No bugs at all and completely efficient and intuitive to use with no need to figure out how to do anything.